Q13 Deeper Project Suggestions

1. **Superposition capacity curve:** extend the module's NumPy demo — sweep N features and sparsity levels at fixed d, plot the phase boundary where readout fails, and fit an empirical capacity law; compare against the Johnson–Lindenstrauss near-orthogonality bound. Report where the toy model agrees and disagrees with Elhage et al.'s phase-change observations.
2. **Probe-vs-use audit:** take a small open model (GPT-2 small or Pythia-1B) and a set of factual relations; train linear probes on mid-layer activations to predict the object token. Then run causal interventions (activation patching of the probed direction) and quantify the gap between "probeable" and "causally used" knowledge.
3. **Mini-ROME on a toy transformer:** train a 2–4 layer transformer on synthetic subject–relation–object triplets; implement causal tracing and a rank-one edit of the mid-MLP matrix; measure paraphrase generalization, neighbor-subject bleedover, and perplexity drift as a function of edit count. Compare single-edit ROME vs. a MEMIT-style batched spread over layers.
4. **Weights vs. context conflict study:** for a set of facts the model knows parametrically, construct prompts whose context contradicts the weights (and vice versa); measure which channel wins as a function of context position, model scale, and fact popularity. Connect the observed behavior to Q12's calibration and confidence questions.